Physics 12 Winter 2026
Week 4 Discussion

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1. **Summary of Topics from Last Week**

- *From the lectures*
  - ◇ Forms of energy: kinetic, potential
    - ∗ All forms of energy we mentioned in class fall into these two categories. Some (such as mechanical) can convert between the two.
  - ◇ Conservation of energy
    - ∗ The examples in lecture mostly focused on mechanical energy - the conversion between kinetic energy and potential energy.
    - ∗ Without dissipative forces, the total energy of an isolated system stays constant.
  - ◇ Dissipative force (friction, air resistance, etc.)
    - ∗ These forces take mechanical (or other types of) energy away from the system by converting it into heat.
  - ◇ Temperature, Internal energy, Heat
    - ∗ Temperature is a measure of average kinetic energy of the random motion of microscopic particles.
  - ◇ States of matter (solid, liquid, gas, plasma)
  - ◇ Change of state (evaporation, boiling, etc.)
    - ∗ Absorbs energy to go from more ordered states to less order states (evaporation). Releases energy to go from less ordered state to more ordered states.
  - ◇ Thermal expansion
    - ∗ Most materials expand when heated and contract when cooled.
  - ◇ Heat capacity, Specific heat, Latent heat
  - ◇ Heat transfer mechanisms
    - ∗ Conduction $P = k\frac{A\Delta T}{d}$. Depends linearly on the difference in temperature.
    - ∗ Convection
    - ∗ Radiation $P = e\sigma AT^4$. Increases fastest relative to change in temperature because of the 4th power.

- ◇ Thermodynamics
  - ∗ Zeroth law: If two systems are both in balance with a third system, they are in balance with each other.
    - · Defines temperature.
    - · If object A is the same temperature as object B, and object B is the same temperature as object C, then A and C are also the same temperature.
  - ∗ First law: $\Delta U = Q - W$. This is just a statement of conservation of energy - the change in internal energy for a system has to come from somewhere in some other form of energy.
  - ∗ Second law: $\Delta S > 0$ (In any physical process, the entropy of the universe always increases).
    - · Energy naturally spreads out and becomes less useful ("You can't break even").
    - · Some consequences: heat always flows from hot to cold; higher quality energy natural tends to become lower quality energy
  - ∗ Third law: As a system approaches absolute zero temperature (0 K or $-273.15°$C), all processes stop and the entropy reaches a minimum.
    - · We can get infinitely close to but never reach the abosolute zero.
- ◇ Heat engine, Heat pump, Thermodynamic efficiency
  - ∗ In a heat engine, we extract a part of the heat flow from a hot reservoir to a cold reservoir into useful work.
  - ∗ Heat pump is just a heat engine in reverse.
- ◇ Entropy
  - ∗ Measures how disordered a system is - for the same substance, gas has higher entropy than liquid, which has higher entropy than solid.
  - ∗ Entropy is related to energy quality. Higher quality energy (electrical, mechanical) tends to naturally transform into lower quality energy (thermal/heat) through dissipative forces.

- *From the textbook* (Wolfson 4th edition)

  There's a lot of overlap between the lectures and the textbook this time!
  - ◇ Thermal Energy, Heat, Temperature (Chapter 4.1 - 4.3)
  - ◇ Heat Capacity, State Changes, Energy Quality (Chapter 4.4 - 4.6)
  - ◇ Entropy, Laws of Thermodynamics, Heat Engine/Pump (Chapter 4.7 - 4.10)

2. ***Phase Change, Heat Capacity and Thermal Transfer**

(a) **The Energy of Ice.** In 2025, the volume of the arctic sea ice hit a new low at around 13000 km^3.

- The latent heat of fusion for water is $L_f = 334$ kJ/kg. How much heat is required to melt 5 kg of ice at 0°C into liquid water at 0°C?

  Solution: We simply multiply the heat of fusion with the mass of the ice

  $$Q = mL_f = 5 \text{ kg} \times 334 \text{ kJ/kg} = 1670 \text{ kJ}. \tag{1}$$

  This is roughly 400 kcal (recall that 1 kcal is about 4 kJ).

- Does the temperature change during this process? If you keep adding heat after the ice is gone, what happens to the molecules?

  Solution: The temperature does not change during a state change (melting), but it will start increasing once the melting has finished.

(b) ***Electric or Gas Stove?**[1] Currently, most of the electricity we use is generated by fossil fuels, of which the majority comes from natural gas, so let's estimate the energy we consume when boiling 1 gallon (0.0038 m^3) of water. Take the specific heat of water to be 4184 J/(kg·K).

- Assume 100% of the heat generated by the stove gets transferred to the water, how much energy (in the form of heat) does the stove need to generate to bring the water from room temperature (20°C) to boiling (100°)?

  Solution: We use the definition of specific heat:

  $$Q = mc\Delta T = 0.0038 \text{ m}^3 \times \frac{10^3 \text{ kg}}{1 \text{ m}^3} \times 4184 \text{ J/(kg} \cdot \text{K)} \times 80 \text{ K} = 1.3 \times 10^6 \text{ J} \tag{2}$$

- Take the heat content of natural gas to be 55 MJ/kg. How much natural gas do we have to burn in a gas stove to generate this much energy?

  Solution:

  $$\frac{1.3 \times 10^6 \text{ J}}{55 \text{ MJ/kg}} = 0.023 \text{ kg} \tag{3}$$

[1] From last week's worksheet

- In a natural gas power plant, roughly a third of the thermal energy from burning natural gas gets converted to electricity. Furthermore, electricity can be very efficiently turned into heat by an electric stove. How much natural gas do we have to burn at the power plant to generate the electricity we need to heat up the same water using an electric stove?

  Solution: Let's assume 100% electricity to heat conversion.

$$\frac{1.3 \times 10^6 \text{ J}}{33\% \times 55 \text{ MJ/kg}} = 0.07 \text{ kg} \tag{4}$$

  It takes more natural gas to do the same thing! This is because the form of energy we need (heat) is a low quality form of energy, whereas electricity is a high quality form of energy. If low quality energy is all we need, we don't have to go through all the trouble to make it high quality and then turn that back into low quality.

(c) **The Three Modes**[2]**.** Match the following scenarios to conduction, convection, or radiation:

- A black asphalt road getting hot on a sunny day: radiation
- Heat escaping through the glass of a single-pane window: conduction
- Warm air rising from a heater to the second floor of a house: convection

(d) ***Heat Loss at Home**[3]**.** The rate of heat conduction through a wall is given by $P = k\frac{A\Delta T}{d}$, where k is the thermal conductivity of the material, ΔT is the difference in temperature across the wall, d and A are the thickness and area of the wall, respectively.

- Let's assume the wall is 0.3 m (about a foot) thick with an area of 10 m^2 and made of concrete ($k = 1$ W/(m·K)). It's winter in San Diego, so the temperature outside is about 10°C while the temperature inside is a comfortable 20°C. Calculate the heat loss through the wall.

  Solution: We use the equation provided:

$$P = k\frac{A\Delta T}{d} = 1 \text{ W/(m} \cdot \text{K)}\frac{10 \text{ m}^2 \times 10 \text{ K}}{0.3 \text{ m}} = 333 \text{ W} \tag{5}$$

- Why is a "dead air space" between two panes of glass a better insulator than a solid sheet of glass?

[2]From last week's worksheet
[3]From last week's worksheet

Solution: Air has a much lower thermal conductivity ($k = 0.026$ W/(m $\cdot$ K)) than glass ($k = 0.8$ W/(m $\cdot$ K)).

- Assume the wall has en emissivity of 1 and has a temperature roughly in between the inside and outside (15°C or 288 K), at what rate does it lose energy through radiation? Use $\sigma = 5.67 \times 10^{-8}$ W/m^2/K^4.

  Solution: The rate of energy loss due to radiation is

  $$P = e\sigma AT^4 = 1 \times 5.67 \times 10^{-8} \text{ W/m}^2/\text{K}^4 \times 10 \text{ m}^2 \times (288 \text{ K})^4 = 3900 \text{ W} \quad (6)$$

  This is a lot! However, while the wall radiates energy away, it also absorbs radiated energy from the surroundings, so the net effect is not as large.

3. **Thermodynamics, Heat Engine and Heat Pump**

   (a) **Internal Energy.** The First Law of Thermodynamics states $\Delta U = Q - W$, where ΔU is the change in internal energy of the system, Q is the energy flowing **into** the system, and W is the work done **by** the system.

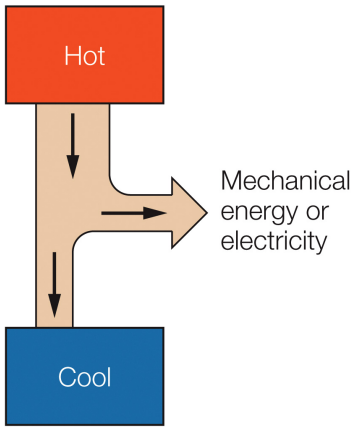
   - If you perform 200 J of work by compressing a gas in a cylinder, and 50 J of heat escapes to the surroundings, what is the change in the gas's internal energy?

     Solution: The work of 200 J is done **on** the gas in the cylinder, so $W = -200$ J. The heat of 50 J flows **out** into the surrounding, so $Q = -50$ J. Therefore, $\Delta U = Q - W = 150$ J.

   - Does the gas get hotter or colder?

     Solution: Let's assume the gas is ideal (which is usually a good assumption), then the temperature increases as the internal energy increases.

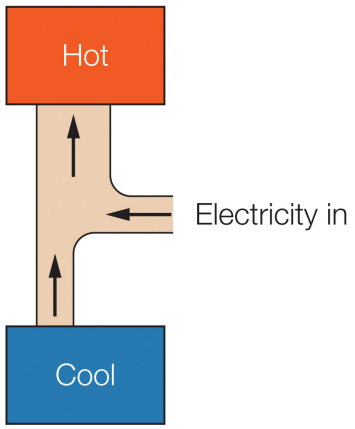
   (b) ***The Carnot Limit.** A hypothetical heat engine takes in heat from a burning fuel and exhausts it into the environment, as depicted by the picture below. A natural gas plant has a combustion temperature of 600°C and exhausts into the air at 25°C. Calculate the maximum theoretical efficiency.



Solution: The maximum theoretical efficiency depends on only the temperatures of the hot and cold reservoirs:

$$e = 1 - \frac{T_c}{T_h} = 1 - \frac{298\text{ K}}{873\text{ K}} = 66\% \tag{7}$$

(c) ***The Heat Pump "Magic"**. A heat pump is used to warm a house. It has a Coefficient of Performance (COP) of 3.5.



- If the house requires 7,000 J of heat to stay warm, how much electrical work must the heat pump consume?

  Solution: The COP is defined as the ratio of the heat delivered to the electrical energy consumed, so we just divide the heat delivered by the COP to get the amount of electrical energy needed: 7000 J/3.5 = 2000 J.

- How much electrical work would a standard "electric toaster" heater (100% efficient) need to provide the same 7,000 J?

  Solution: It just converts electrical energy to heat at 100% efficiency, so 7000 J.